

Hany Elgala — Curriculum Vitae

Electrical and Computer Engineering 301C CNSE Downtown

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EMPLOYMENT

2022–present Associate Professor, University at Albany, Albany, NY, USA
2015–2022 Assistant Professor, University at Albany, Albany, NY, USA
2014–2015 Research Professor, Boston University, Boston, MA, USA

ADMINISTRATIVE AND LEADERSHIP POSITIONS

Fall 2025–present Acting Director, AI & Society College, University at Albany, Albany, NY, USA
Spring 2025 Associate Director, AI & Society College, University at Albany, Albany, NY, USA
2024–2025 Provost Fellow, AI and Academic Integrity, University at Albany, Albany, NY, USA
2024–2025 Member, Leadership Academy, University at Albany, Albany, NY, USA
2023–present Program Director, M.S. in Electrical and Computer Engineering, Albany, NY, USA

HONORS, AWARDS, FELLOWSHIPS, AND AFFILIATIONS

Spring 2026 Catalyst Award, College of Nanotechnology, Science, and Engineering Honors Convocation
Winter 2025 TÜBİTAK Visiting Scientist Fellow, Scientific and Technological Research Council of Türkiye
2024 Best Paper Award, IEEE Middle East Conference on Communications & Networking MECOM
2023 Included in the Stanford University/Elsevier list of the world's top 2% of scientists
2022–2024 Humboldt Experienced Researcher Fellow, Alexander von Humboldt Foundation, Germany
2022–present Fellow, Center for Technology in Government, University at Albany, USA

OTHER PROFESSIONAL APPOINTMENTS

Winter 2025–2026 Visiting Professor, Özyeğin University, Istanbul, Türkiye
Summer 2022–2025 Visiting Professor, TU-Berlin, Fraunhofer Heinrich-Hertz-Institute, Berlin, Germany
Winter 2019–2020 Visiting Professor, Khalifa University, Abu Dhabi, United Arab Emirates
Fall 2014 (part-time) Lecturer, Computer Networks, Boston University, Boston, MA, USA
2003–2012 Residential College Supervisor & Interim College Master, Constructor University*, Germany
Spring 2001 (part-time) Lecturer, Obour Institute for Engineering and Technology, Egypt

PUBLICATIONS

Google Scholar metrics as of July 7, 2026 (see complete publication list below)

	All	Since 2021
Citations	9349	3323
h-index	38	28
i10-index	73	51

EDUCATION

2012–2014 Senior Postdoctoral Researcher, Boston University, Boston, MA, USA
2010–2012 Postdoctoral Researcher, Constructor University, Bremen, Germany
2010 Ph.D., Constructor University, Bremen, Germany
2003 M.S., Furtwangen University, Germany
2000 B.S., Ain Shams University, Egypt

ADDITIONAL EMPLOYMENT

2003–2006 Laboratory Engineer, Constructor University, Bremen, Germany
Sep. 2002–Jun. 2003 Researcher, Infineon Technologies, Munich, Germany
Jan.–Sep. 2001 Satellite Engineer, NILESAT, Cairo, Egypt
Jun.–Dec. 2000 Network Services Engineer, Alcatel-Lucent, Cairo, Egypt

* Formerly International University Bremen and Jacobs University Bremen

AI & SOCIETY COLLEGE PROJECTS, PROGRAMS, AND EVENTS

Projects

- Empire AI/SUNY Partnership: Advancing AI for the Public Good Initiative
- SUNY Academic Innovation Grant: AI & Society Commons: A SUNY Digital Knowledge Lab for Teaching, Learning, and Critical Inquiry

Programs

- Faculty Academic Innovation Fellowship
- Doctoral Dissertation Fellowship
- Master's Experiential Learning Program
- Discipline Specific 7-Credits Micro-credential
- UUNI 118: Introduction to Artificial Intelligence Course
- AI Preparedness Academy
- Faculty Exchange and Affiliation Program
- AI Makerspace
- Partnership with Information Technology Services (ITS) to develop and host AI tools and Apps.

Events

- AI Plus Symposium: Future of Education
- AI & Society College Fellows Program at Showcase Day
- AI & Society College Social
- AI for Good Challenge
- Workshops at the Albany City Library

SERVICE

University Service

- Member, Review Committee, UAlbany Chancellor's Dissertation Award Nomination Committee, Summer 2026.
- Member, Steering Committee, President's AI Task Force, 2026.
- Member, Teaching and Learning Committee (AI Literacy Subcommittee), President's AI Task Force, 2026.
- Member, Research Committee, President's AI Task Force, 2026.
- Panelist, PEF Statewide ITS AI Summit, Albany, NY, June 2, 2026.
- Panelist, "AI for the Public Good," moderated by New York State Assemblymember Nily Rozic, NYS AAPI Summit, Albany, NY, May 8th, 2026.
- Panelist, North African and Arab Society (NAAS), "Voices for a Cause": NAAS Panel & Fundraiser, April 28th, 2026.
- Co-chair, Organizing Committee, AI Plus Symposium, March 6th, 2026.
- Panelist, North African and Arab Society (NAAS) Professional Event, November 20th, 2025
- Panel moderator, "AI in Higher Education," First-Year Experience Conference, 2025.
- Chair, AI Plus Institute Research Group, Humanitarian Aid, Public Policy and Education, 2024-2025.
- Presenter, "AI and Ethics," GradEdge Workshop Series, The Graduate School, 2024-2025.
- Volunteer and attendee, University-wide Undergraduate Commencement, 2025 and 2026.
- Member, Undergraduate Student Commencement Speaker Committee, 2024-2026.
- Participant and Mentor, UAlbany Showcase Day, including undergraduate research, capstone teams, and course projects, 2024-2026.
- Member, AI Plus Curriculum Committee, 2024-2025.
- Member, University Senate; University Planning and Policy Council; and University Facilities Committee, 2022-2024.
- Presenter, University Undergraduate Research Fair, Fall 2017.
- Member, International Education Advisory Committee, 2017-2019.
- Reviewer, Faculty Research Award Program (FRAP-A), 2015-2016.
- Faculty participant, lunch with first-year students on the Computer Engineering major, 2015.
- Faculty participant, dinner with Phi Beta Kappa Visiting Scholar David Campbell, 2015.

SUNY Service

- Member, Policy Committee, SUNY INSPIRE Center, a system-wide hub for AI research, scholarship, and workforce development, 2026 (nominated).
- Contributor to the joint SUNY COIL/UAS7 proposal to the German Academic Exchange Service (DAAD); €250,000 awarded, 2020-2021.

College Service

- Name reader, Undergraduate and Graduate Commencement Stage Crossing, 2025 and 2026.
- Committee member, development of a dual M.S. program between UAlbany and Universitat Politècnica de València (Spain) in AI and Energy Systems, 2023-present.
- Member, AI Plus Initiative Faculty Cluster Hire Committee, 2023.
- Faculty participant, Accepted Student Open House, 2023 and 2024.
- Faculty recruiter, FIRST Robotics, MVP Arena, Albany, NY, March 31-April 1, 2023.
- Member, College Commencement Speaker Committee, 2019.

- Member, Teaching Assistant Guidelines Committee, 2017.
- Member, Smart Cities Grant Committee, in collaboration with the City of Schenectady, NY, 2017.

Department Service

- Member, Student Academic Grievance Committee, 2025–present.
- Program Director, M.S. in Electrical and Computer Engineering, 2023–present.
- Member and Chair, Graduate Curriculum Committee, 2015–2023.
- Member, Doctoral Candidacy Examination Committee, 2019–present.
- Member, Graduate Admissions Committee, 2019.
- Member, Doctoral Qualifying Examination Committee, 2018–present.
- Member, Department Chair Search Committee, 2016.
- Coordinator, student visit to GlobalFoundries in Malta, NY, including a lecture by 2014 Nobel Laureate in Physics Shuji Nakamura, 2016.
- Department representative, Schuyler Building renovation planning meetings, 2016.
- Member, Administrative Manager Search Committee, 2016.
- Department representative, Open House, Fall 2015.
- Member, Faculty Search Committee, 2015.

ACADEMIC LEADERSHIP POSITIONS

- Member, OFC 2027 Organizing Committee, Subcommittee S5, Los Angeles, CA, USA; appointed February 2026.
- General Chair, IEEE International Conference on Future and Intelligent Communication and Networking (FICN), San Antonio, TX, USA, December 8–10, 2026.
- Editor-in-Chief, SCIFINITI Communications & Networks Connect, 2024–present.
- Associate Editor, Optical Communications and Networks section, Frontiers in Communications and Networks, 2020–present.
- Editor, Optical Wireless Systems and Networks, IEEE Transactions on Communications, 2017–2022.
- Topic Editor, “Coexisting Radio and Optical Wireless Deployments (CROWD),” Optical Communications and Networks section, Frontiers in Communications and Networks, 2021.
- Program Chair, Signal Processing in Photonic Communications (SPPCom), Advanced Photonics Congress, 2020 and 2021.
- Lead Editor, special issue on “Coexisting Radio and Optical Wireless Deployments,” IEEE Transactions on Cognitive Communications and Networking, published December 2019.
- Guest Editor, special section on “Optical Wireless Technologies for 5G Communications and Beyond,” IEEE Access, 2017.
- Symposium Chair, Wireless Optical Networks, International Wireless Communications and Mobile Computing Conference, 2018.
- Contributor, IEEE Communications Society Best Readings in Optical Wireless Communications, published August 2018.
- Communications Testbed Leader, NSF Lighting Enabled Systems & Applications Engineering Research Center, 2015–2016.

TEACHING

University at Albany, USA (2015–present)

- Introduction to Machine Learning for Engineers (IECE 465/565; formerly IECE 494/560), undergraduate and graduate, Spring 2024–2026.
- Advanced Digital Communications (IECE 472/572), undergraduate and graduate, Fall 2022 and 2023.
- Intelligent Internet of Things (IECE 494/560), new undergraduate/graduate course with a **COIL component in partnership with Osnabrück University of Applied Sciences**, Spring 2020, 2021, and 2023.
- Digital Signal Processing (IECE 462/562), undergraduate and graduate, Spring 2020 and 2021.
- Digital Systems and Laboratory (IECE 231), undergraduate, Fall 2019–2021 and Spring 2022 and 2024.
- Signals and Systems (ICEN 340), undergraduate, Spring 2016, 2018, and 2019; Fall 2018.
- Digital Logic Design (ICEN 340), undergraduate, Fall 2015–2017.
- Independent Research in Computer Engineering (ICEN 497), undergraduate.
- Independent Study and Research (ICSI 890), graduate.

Boston University, USA (2014)

- Introduction to Computer Networking (EC 441), undergraduate, Fall 2014.

Constructor University, Germany (2003–2006)

- Signals and Systems Laboratory (300221), undergraduate.
- Digital Signal Processing Laboratory (300221), undergraduate.
- Electronics Laboratory (300222), undergraduate.
- Guided Research in Electrical Engineering and Thesis (300342), undergraduate.
- Guided Research in Electrical Engineering (300361), undergraduate.

Obour High Institute for Engineering and Technology, Egypt (2001)

- Electronics Laboratory, undergraduate.
- Physics, undergraduate.
- C Programming, undergraduate.

OUTREACH ACTIVITIES

University at Albany, USA

- Instructor, one-day “Introduction to AI” session for high school girls, in collaboration with the Girls Inc. Eureka! Program and UAlbany College of Arts and Sciences, July 22, 2026.
- Instructor, two-weekend AI program with RiseHIGH in Schenectady, NY, April 5 and 12, 2025. Introduced students in grades 7–12 to classical machine learning, neural networks, responsible generative AI, model training, and prompt engineering through hands-on activities.
- Workshop leader, “Connecting the Unconnected,” Science & Technology Entry Program, for students in grades 7–10, July 2020.
- Instructor, Science & Technology Entry Program Robotics Team, UAlbany, Fall 2019–2020. Mentor students in grades 7–12 in analytical thinking, teamwork, problem-solving, and VEX robot design and programming for local and national competitions.
- Instructor, one-week “IoT and Embedded Systems” session for high school girls, in collaboration with the Girls Inc. Eureka! Program and UAlbany College of Arts and Sciences, Summer 2018.
- Faculty participant, one-day College of Engineering and Applied Sciences visit to Hackett Middle School, 2017.

Boston University, USA

- Research mentor, Research Internship in Science and Engineering program, for high school student Eric D’sa; project: “Using Solar Cells in Energy-Harvesting Optical Transceivers,” six weeks, Summer 2014.
- Instructor, Summer Challenge in Electrical Engineering, for 20 high school students; 10 lectures, one tutorial, and eight experiments over two weeks, Summer 2013.

GRANT FUNDING

Industrial

- Principal Investigator, IBM–UAlbany Center for Emerging Artificial Intelligence Systems, Theme II: “Deep Learning to Coexist: AIU-Accelerated Multi-Task Wireless Waveforms”; \$117,000 total; awarded; project began January 2026.

Federal

- Co-Principal Investigator, National Science Foundation, Major Research Instrumentation Track 1: “Acquisition of InSpect: Instrument for Research and Education in Spectrum Science”; \$1,300,000 total; awarded; project began May 2026.
- Co-Principal Investigator, National Science Foundation, CRI: II-NEW: “CHRONOS—A Cloud-Based Hybrid RF-Optical Network over Synchronous Links”; \$749,999 total; awarded; September 2018–August 2021, extended through September 2024.
- Principal Investigator, National Science Foundation, NeTS Small Collaborative Research: “Coexistence of Directional Communications within 5G Networks: The Case of Visible-Light-Enhanced Small Cells,” with Boston University and New Jersey Institute of Technology; \$515,613 total, including \$154,787 to UAlbany; awarded; September 2016–September 2020.
- Principal Investigator, National Science Foundation Engineering Research Center for Lighting Enabled Systems & Applications: “Illumination, Sensing, and Communication in OFDM-Based Visible Light Communication Systems,” with Boston University and Rensselaer Polytechnic Institute; \$32,729,057 total center award, including an \$89,307 UAlbany subaward; January 2015–August 2018.

State and SUNY

- Principal Investigator, SUNY Academic Innovation Grant, “AI & Society Commons: A SUNY Digital Knowledge Lab for Teaching, Learning, and Critical Inquiry”; \$40,000 total; awarded; June 2026–June 2028.
- Principal Investigator, SUNY AI Partnership/Empire AI, “Advancing AI for the Public Good”; \$900,000 total; awarded; January 2026–June 2028.
- Co-Principal Investigator, U.S. Department of Labor proposal, “The AI-Ready Educator Initiative”; \$3,985,355 requested; submitted December 1, 2025; not funded.

Foundations and Other External Sponsors

- 2221 - Fellowships Grant, TÜBİTAK Visiting Scientist Fellow, Scientific and Technological Research Council of Türkiye; approximately \$5,000 total; awarded Winter 2025–2026.
- Co-Principal Investigator, Abu Dhabi Award for Research Excellence, “Indoor Hybrid Flying Networks for Safety Applications and Services (FlyServ),” with Al Ain University and Khalifa University; approximately \$272,000 total; awarded; 2020–2024.
- Principal Investigator and Humboldt Research Fellow for Experienced Researchers, Alexander von Humboldt Foundation, Germany; approximately \$100,000; awarded; 2022–2024.

- Principal Investigator, National Science Foundation I-Corps technology-transfer award; \$3,000; awarded; Spring 2019.
- Principal Investigator, Metropolitan Transportation Authority state contract, "Intelligent Transportation-Based UWB Positioning and Connectivity"; \$80,969 total and UAlbany share; awarded; November 2016–August 2017.
- Principal Investigator, Bremen Funding Program (BRUT), European Regional Development Fund, Germany; €5,000; awarded; Spring 2012.

University at Albany

- UAlbany Innovation Scholarship for technology transfer; \$1,200; awarded; Spring 2019.
- Third Place, UAlbany Shark Tank technology-transfer competition; \$1,000; Spring 2019.
- Principal Investigator, Presidential Innovation Fund for Research and Scholarship proposal, "Li-Emo: Light-Based Emotion Recognition toward a Human-Centered Internet of Things"; \$55,000 requested; submitted September 2016; not funded.

PROFESSIONAL SERVICE

Invited Talks

- "AI & Ethics: Opportunities and Responsibilities," Graduate School, University at Albany, 2026.
- "AI & Society College Initiatives," Speakers Program, University at Albany Emeritus Center, October 20, 2025.
- "AI & Ethics: Opportunities and Responsibilities," Graduate School, University at Albany, September 25, 2025.
- "Optical Wireless Communication: Applications, Challenges, and Competition," IEEE Radio Frequency Integrated Circuits Symposium (RFIC), June 2025.
- "High-Altitude Platform Stations (HAPS) for Reliable Connectivity and Navigation Services," webinar organized by the Center for Information and Security Education and Research (CINSER) and the Center for Assured and Resilient Navigation in Advanced Transportation Systems (CARNATIONS), March 15, 2025.
- "Augmented Waveforms for Next-G Wireless Networks," EDGE Symposium: Inventing the Future with Advances in Edge, 5G, Controls, Cybersecurity, AI, and Robotics, General Electric, Niskayuna, New York, September 2023.
- "Integration of Services Using New LiFi Waveforms," MAI Colloquium, Fraunhofer HHI, Metro, Access and In-House Systems Group, Berlin, Germany, August 2022.
- "Augmented Waveforms for Future Wireless Networks," Department Seminar, Technische Universität Berlin, Berlin, Germany, July 2022.
- "Augmented Waveforms for Future Wireless Networks," University Seminar, Al Ain University, Al Ain, United Arab Emirates, February 2022.
- "Visible Light Communications for 5G and Beyond," Optica CareerLab webinar, September 2020.
- "Visible Light Communications: From Classical to Machine-Learning Links," IEEE Schenectady Section Symposium Series, Albany, New York, November 2019.
- "Visible-Light-Based Backscatter Communication," Optical Wireless Communication Workshop, European Conference on Optical Communication, September 2017.
- "Flavors of Optical OFDM and Their Dimming Compatibility," Visible Light Communications Workshop, International Conference on Information Optics and Photonics, July 2016.
- Invited presentation, IEEE Wireless Communications and Networking Conference, April 2016.
- "Get Connected Through Light," Helen and John C. Hartmann Department of Electrical and Computer Engineering Seminar, New Jersey Institute of Technology, April 2016.
- "Lighting in the Information Age," Solid-State Lighting Ecosystem, September 2014.
- Invited presentation, New England Workshop on Software Defined Radio, June 2014.
- "Wireless Communications Potential of Solid-State Lighting: Visible-Light Communications," Advanced Energy Conference, April 2013.
- "High-Speed Wireless Networking Using Indoor LED Lighting," Department of Electrical and Computer Engineering Seminar, Boston University, November 2013.
- "A Visible Approach to the Spectrum Crunch," Department of Electrical Engineering and Computer Science Colloquium, Tufts University, February 2013.

Panels

- Panelist representing the University at Albany and the AI & Society College, PEF Statewide ITS AI Summit, Albany, NY, June 2, 2026.
- Panelist representing the University at Albany and the AI Plus Initiative, "AI for the Public Good," moderated by New York State Assemblymember Nily Rozic, NYS AAPI Summit, Albany, NY, May 8th, 2026.
- Panelist, North African and Arab Society (NAAS), "Voices for a Cause" – NAAS Panel & Fundraiser, April 28th, 2026.
- Panelist, North African and Arab Society (NAAS) Professional Event, November 20th, 2025.
- Panel Moderator, "Generative AI and First-Year Teaching," First-Year Experience Conference, May 15, 2025.
- Panelist, "Electromagnetic Hypersensitivity," School of Public Health, University at Albany, virtual, February 2022.
- Panelist, "Teaching and the Interface of Machine Learning in Engineering and Finance," Global Finance Conference, virtual, May 2021.

Event Organization

- Member, OFC 2027 Organizing Committee, Subcommittee S5, Los Angeles, CA, USA; appointed February 2026.

- General Chair, IEEE International Conference on Future and Intelligent Communication and Networking (FICN), San Antonio, TX, USA, December 8–10, 2026.
- Organizer and panelist, “Teaching and the Interface of Machine Learning in Engineering and Finance,” 28th Annual Global Finance Conference, May 27–29, 2021; organized with Rita Biswas, School of Business, University at Albany.
- Organizer and facilitator, inaugural online webinar and panel on Coexisting Radio and Optical Wireless Deployments (CROWD), February 28, 2020.
- Organizer, CROWD Workshop, IEEE Wireless Communications and Networking Conference, Marrakech, Morocco, April 15–18, 2019.
- Organizer, CROWD Workshop, IEEE International Conference on Network Protocols, Cambridge, United Kingdom, September 24–27, 2018.
- Organizer, CROWD Workshop, IEEE International Symposium on Personal, Indoor and Mobile Radio Communications, Montreal, Canada, October 8–13, 2017.

Session Chairing

- “Sensing and Communications,” IEEE Middle East Conference on Communications and Networking, Cairo, Egypt, November 6, 2025.
- “Nonlinear Processing and Quantum Networks,” Signal Processing in Photonic Communications, Advanced Photonics Congress, 2021.
- “Microwave Photonics and Comb Generation,” Signal Processing in Photonic Communications, Advanced Photonics Congress, 2021.
- “Transmission Experiments,” Signal Processing in Photonic Communications, Advanced Photonics Congress, 2020.
- “Signal Processing II,” Signal Processing in Photonic Communications, Advanced Photonics Congress, 2020.
- “Signal Processing I,” Signal Processing in Photonic Communications, Advanced Photonics Congress, 2019.
- “Free-Space Optics and Visible Light Communication,” Signal Processing in Photonic Communications, Advanced Photonics Congress, 2019.
- “Visible Light Communications II,” IEEE Global Communications Conference, 2018.
- “Access Networks and Free-Space Communications,” Signal Processing in Photonic Communications, Advanced Photonics Congress, 2018.
- “Wireless Communications: Fundamentals and PHY I,” IEEE Consumer Communications and Networking Conference, 2018.
- “Physical Layer PHY-02,” IEEE Wireless Communications and Networking Conference, 2015.
- “Physical Layer PHY-15,” IEEE Wireless Communications and Networking Conference, 2015.

Grant Review

- National Science Foundation: CISE (2019), ENG (2020), and NeTS-Medium (2024), United States.
- Natural Sciences and Engineering Research Council of Canada, Discovery Grants Program, December 2022.
- U.S. Department of Energy, Advanced Scientific Computing Research, 5G-Enabled Energy Innovation: Advanced Wireless Networks for Science, 2021.
- Fulbright Senior Award, 2019 and 2020–2021, United States.
- United Arab Emirates University, 2017.
- King Abdullah University of Science and Technology, Competitive Research Grants Program, 2016, Saudi Arabia.
- Dutch Research Council (NWO), 2015, Netherlands.

Technical Program Committees

- Machine Learning and AI for Communications Track, IEEE 95th Vehicular Technology Conference (VTC2022-Spring), Helsinki, Finland, June 19–22, 2022.
- Optical Wireless Communication Workshop, IEEE Global Communications Conference, San Diego, California, December 6, 2015.
- Optical Networks and Systems Symposium, IEEE Global Communications Conference, San Diego, California, December 6–10, 2015.
- Workshop on Visible Light Communications and Networking, IEEE International Conference on Communications, London, United Kingdom, June 8–12, 2015.
- IEEE Vehicular Technology Conference (VTC2015-Spring), Glasgow, Scotland, May 11–14, 2015.
- IEEE Wireless Communications special issue on Visible Light Communications, April 2015.
- IEEE Workshop on Optical Wireless Communications, co-located with IEEE GLOBECOM, Austin, Texas, December 8–12, 2014.
- Optical Wireless Communications Symposium, IEEE Conference on Wireless Communications and Signal Processing, Hefei, China, October 23–25, 2014.
- IEEE Region 10 Symposium (TENSYP), Kuala Lumpur, Malaysia, April 14–16, 2014.
- IEEE Workshop on Optical Wireless Communications, co-located with IEEE GLOBECOM, Atlanta, Georgia, December 9–13, 2013.
- International Conference on Connected Vehicles and Expo, Las Vegas, Nevada, December 2–6, 2013.
- IEEE International Conference on Photonics, Melaka, Malaysia, October 28–30, 2013.
- International Workshop on Fiber Optics in Access Networks, Almaty, Kazakhstan, September 11–12, 2013.

- IEEE Workshop on Optical Wireless Communications, co-located with IEEE GLOBECOM, Anaheim, California, December 3–7, 2012.

Selected Peer-Review Service

- IEEE: Journal of Lightwave Technology; Journal of Optical Communications and Networking; IEEE Photonics Journal; IEEE Photonics Technology Letters; IEEE Communications Letters; IEEE Transactions on Communications; IEEE Transactions on Signal Processing; IEEE Wireless Communications; and conferences including GLOBECOM, ICC, PIMRC, and VTC.
- Optica: Optics Express; Optics Letters; and Chinese Optics Letters.
- The Royal Society: Philosophical Transactions of the Royal Society A.
- Institution of Engineering and Technology: IET Communications.
- Elsevier: Digital Signal Processing and Optics Communications.
- EURASIP Journal on Advances in Signal Processing.

Webinars

- “Visible Light Communications for 5G and Beyond,” Optica CareerLab 5G Webinar Series, September 28, 2020.

Tutorials

- Tutorial, IEEE Wireless Communications and Networking Conference, April 2016.

Community Service

- Contributing expert, “How Should the MTA Update Its Signal Technology?” City & State New York, April 2019.

Conference Presentations

- Oral presentation, IEEE Middle East Conference on Communications and Networking, Cairo, Egypt, November 4–6, 2025.
- Poster presentation, Northeast Regional Defense Technology Hub Workforce Development Summit, Albany NanoTech Complex, Albany, New York, October 2, 2025.
- Oral presentation, IEEE Middle East Conference on Communications and Networking, Abu Dhabi, United Arab Emirates, November 17–20, 2024.
- Oral presentation, “Digital Signal Processing I,” Signal Processing in Photonic Communications, Optica Advanced Photonics Congress, August 2019.
- Oral presentation, “Optical Network Architectures,” IEEE Global Communications Conference, December 2018.
- Oral presentation, Asilomar Conference on Signals, Systems, and Computers, November 2014.
- Oral presentation, IEEE International Symposium on Personal, Indoor and Mobile Radio Communications, September 2014.
- Oral presentation, Optical Fiber Communication Conference and Exposition, March 2014.
- Oral presentation, IEEE International Symposium on Personal, Indoor and Mobile Radio Communications, September 2010.
- Oral presentation, Wireless and Optical Communications Networks Conference, April 2009.
- Oral presentation, IEEE Vehicular Technology Conference, April 2007.

OTHER SCHOLARLY ACTIVITY

Artistic Exhibitions and Performances

- Participated in the “Tech, Drugs, and Rock n’ Roll” networking event, Boston, Massachusetts, July 15, 2014.
- Exhibited at “Highlights of Physics,” Congress Centre Bremen, Bremen, Germany, October 6–10, 2006.
- Demonstrated visible light communication at Information Night, House of Science, Bremen, Germany, June 14, 2006.

Consultancies

- Paxeramed Corp., Newton, Massachusetts, USA, June 2013–June 2014.
- University of Tabuk, Tabuk, Saudi Arabia, March 2012–March 2013.

Editorial Commentary and Media Coverage

- Research on intelligent-transportation applications of ultra-wideband positioning was featured in The New York Times and Metro Magazine.

PROFESSIONAL AFFILIATIONS

Memberships

- IEEE, 2008–present.
- Optica (formerly the Optical Society of America), 2014–present.
- SPIE, 2010–2011.

STUDENT SUPERVISION

Ph.D. Supervision

University at Albany, USA

- Honglan Chen, Ph.D. student, Fall 2024–present; expected graduation: Fall 2029.

- Saksham Dewan, Ph.D. student, Fall 2023–present; expected graduation: Fall 2028.
- Monette Khadr, Ph.D., 2017–2021. Dissertation: “Augmented Communications: A Solution for Overcoming High Spatial Correlation of the Massive-MISO VLC Channel.” Graduated Fall 2021.
- Ahmed Hussein, Ph.D., 2016–2021. Dissertation: “Mixed-Carrier Communication for Sixth-Generation Networks.” Graduated Spring 2021.

Ph.D. Co-Supervision

Boston University, USA

- Yuting Zhang, 2014 – 2015, Lighting optimization with user activity recognition in smart rooms
- Pankil Butala, 2013 – 2015, MIMO techniques in visible light communication

Constructor University, Germany

- Irina Stefan, 2010 – 2013, Multiuser access and link adaptation techniques for indoor optical wireless communication systems
- Crisanto Quintana Sánchez, PhD exchange student (Spain), July – October 2010, Experimental measurements for hard-decoded and soft-decoded IM/DD OFDM optical wireless links

Ph.D. Thesis Committee Membership

- John W. Bradburn, University at Albany, Doctoral Candidacy Examination Committee, June 2024.
- Imara Mohamed Nazar, University at Albany, Dissertation Committee, April 2024.
- Marco Schaarschmidt, Osnabrück University, Germany, Dissertation Committee, October 2023.
- Hesham Mohammad, University at Albany, Dissertation Committee, July 2023.
- Maqsood Abdul Careem, University at Albany, Dissertation Committee, May 2023.
- Shuvam Chakraborty, University at Albany, Doctoral Candidacy Examination Committee, November 2022.
- Habibzadeh Hadi, University at Albany, Dissertation Committee, October 2022.
- Hesham Mohammad, University at Albany, Doctoral Candidacy Examination Committee, Spring 2022.
- Habibzadeh Hadi, University at Albany, Doctoral Candidacy Examination Committee, Spring 2021.
- Ehab Al Badawy, University at Albany, Doctoral Candidacy Examination Committee, August 2020.
- Wei Xiong, University at Albany, Dissertation Committee, May 2020.
- Sihua Shao, New Jersey Institute of Technology, Dissertation Committee, April 2018.
- Jimmy Chau, Boston University, Dissertation Committee, June 2016.
- Michael Rahim, Boston University, Dissertation Committee, August 2015.

Master's Student Supervision

University at Albany, USA

- Saksham Dewan, M.S. thesis, “Multi-Task Learning for Hybrid Waveforms,” April 2024.
- Sai Sandeep Kajukuri, M.S. project, “Multi-Task Learning for Signal and Modulation Classification,” December 2022.
- Saksham Dewan, independent research, “Using Multi-Task Learning for Modulation Classification and Direction-of-Arrival Estimation,” Fall 2022.
- Priti Pachpande, M.S. thesis, “Communications Using Deep Learning Techniques,” May 2019.
- Vasuda Trehan, independent research project, “Wearable Devices That Keep You Safe,” Spring 2018.

Boston University, USA

- Emily Lam, 2014 – 2015, Spectral and energy efficient optical OFDM formats
- Vipul Agarwal, Student Assistant, 2014, Network enabled light fixtures
- Bowen Song, Student Assistant, 2013, Impedance matching of LEDs for Gbps transmission
- John Gancarz, thesis, 2013, Energy efficiency and light quality control in VLC systems

Constructor University, Germany

- Ahmad Sadjadpour, student assistant, FPGA implementation of sampling time-offset estimation in optical OFDM systems.
- Dipesh Mistry, Student Assistant, 2011, Ethernet video streaming for optical wireless OFDM demonstrator
- Ahmad Sadjadpour, Student Assistant, 2011, FPGA control of a voltage gain amplifier for optical wireless OFDM demonstrator
- Marwan Mammouda, semester project, 2010, Link budget of Mbps optical wireless indoor communication
- Rashid Mehmood, semester project, 2009, Optical channel modeling and optical spatial modulation
- Irina Stefan, thesis, 2009, FPGA implementation of an optical wireless communication system based on OFDM
- Svilen Dimitrov, thesis, 2009, SINR and path-loss modeling of a cellular optical wireless system in aircraft cabin

Master's Thesis Committee Membership

- Saksham Dewan, University at Albany, committee chair, April 2024.
- Marc S. DiRaimo, University at Albany, committee member, November 2023.
- Shuvam Chakraborty, University at Albany, committee member, May 2022.

Undergraduate Student Supervision and Mentoring

University at Albany, USA

- Sanad Sahawneh and Brian Allen: UG Capstone Project 2025-2026, Low-Cost RF Exposure Monitoring with TinyML Prediction.
- Marianna Karagiannis, Joshua Muhammad, and Faiyaz Sajjad: UG Capstone Project 2025-2026, developing an open-source LiFi system using Microprocessor Boards.
- Justin Adams, Jeremy Allen, Anna Beech, and Hasan Merzai, Team ULTRA (U-Albany LiFi Team with Real-World Applications): UG Capstone Project 2024-2025, developing an open-source LiFi system using FPGA and PoE-enabled. First Place ECE research at the UAlbany Showcase Day 2025.
- Hemel Debnath, Tony J Leggiero, Sarah Scheps, Population RF Exposure Team: UG Capstone Project 2024-2025, building a cloud-enabled RF exposure sensor network. Presented at the UAlbany Showcase Day 2025.
- Kristjan Bruno, Cody Davies, Michael Jacques, and Kihyun Lee, UG Capstone Project 2023-2024, Low-Cost Arduino-Based Radio Frequency Power Monitoring Sensor Network. First Place ECE research at the UAlbany Showcase Day 2024.
- Basit Bolach, Dual Communications using VLV and RF leakage, spring 2024. Presented at the UAlbany Showcase Day 2024.
- Xiuxian Wu, VLC Communications using commercial LED panels, fall 2024.
- David P Kiszka, Creating lab manuals for the CAROL edge TPU, fall 2023.
- Eric Dowd, UWB signal propagation using different GUIs, spring 2022.
- Anya Ross, Autoencoder Model for OFDM-Based Optical Wireless Communications, spring 2020 – spring 2021. Presented at the UAlbany Undergraduate Research Conference 2021. The 2021 Presidential Award for Undergraduate Research at the University at Albany.
- Walter, Ian, 1) Deep learning based optical camera communications and 2) Massive augmented spatial modulation using machine learning, spring 2019 – spring 2020. The 2020 Presidential Award for Undergraduate Research at the University at Albany.
- Jake Eidelkind, Augmented reality (AR) prototype using VLC, spring 2019 – present
- Orsi Beke and Vincente Bruno, IR-based motion capture system for autonomous indoor UAVs, spring and fall 2019
- Allanah Matthews and Ambrose Ngayinoko, VLC-enabled LED screen panels, spring 2019
- Robert Kats and John W Bradburn, VLC-enabled robots, fall 2018
- Travis Cooper and Ryan Abuasi, Building a passive VLC transceiver, fall 2018
- Meng Zhang and Na Li, IR-based motion capture system for autonomous indoor UAVs, fall 2018
- Martin M Andreev, VLC-enabled LED screen panels, fall 2018
- Patrik Fowley, Energy harvesting for wearable devices and body area networks (BANs), fall 2017
- Elaine Huang, organic LEDs for communication and ranging, fall 2016- fall 2018. Presented at the annual University at Albany Undergraduate Research Conference, and the SUNY Undergraduate Research Conference at SUNY Cobleskill. Attended the Microsoft Indoor localization competition.
- Melanie Borgeest, IR-based motion capture system for autonomous indoor UAVs, spring 2017

Boston University, USA

- John Glynn, Software-defined radio-based demonstration of optical OFDM, semester project, 2014
- Emily Lam, Beam tracking based on motion capture and DMX-over-Ethernet, semester project, 2014
- Jenesis Jones, OPPM implementation based on Arduino, semester project, 2014

Constructor University, Germany

- Dipesh Mistry, Development of algorithms for Ethernet communication between Altera FPGA and user PC, thesis, 2009
- Mihai Voica, Graphical user interface development using Altera NIOS II tool and Sharp C, thesis, 2009
- Dipesh Mistry, Performance of multi-carrier modulations for indoor optical wireless communication systems, semester project, 2009
- Nikolay Melnikov, Implementation of optical OFDM system on the Texas Instruments C6713 DSP platform, thesis, 2007
- Bogdan Pricope, Visible light communication using OFDM and Simulink toolboxes, thesis, 2006
- Mostafa Afgani, C-implementation of optical OFDM transmitter on the TI C6711 DSP platform, semester project, 2005
- Patrick Agyapong, C-implementation of optical OFDM receiver on the TI C6711 DSP platform, semester project, 2005
- Ellina Foutekova, C-implementation of optical OFDM synchronization and equalization on the TI C6711 DSP platform, semester project, 2005
- Vinod Patmanathan, Simulation and DSP board implementation of an optical transmission system using OFDM, thesis, 2004

Student Awards

- Justin Adams, Jeremy Allen, Anna Beech, and Hasan Merzai, Team ULTRA (U-Albany LiFi Team with Real-World Applications): UG Capstone Project 2024-2025, developing an open-source LiFi system using FPGA and PoE-enabled. First Place ECE research at the UAlbany Showcase Day 2025.
- Kristjan Bruno, Cody Davies, Michael Jacques, and Kihyun Lee, UG Capstone Project 2023-2024, Low-Cost Arduino-Based Radio Frequency Power Monitoring Sensor Network. First Place ECE research at the UAlbany Showcase Day 2024.

- Anya Ross, BSc Student, the 2021 Presidential Award for Undergraduate Research at the University at Albany. Project titled: Autoencoder Model for OFDM-Based Optical Wireless Communications Systems Employing Augmented Communications. Accepted publication at the OSA Advanced Photonics Congress 2021.
- Walter, Ian, BSc Student, the 2020 Presidential Award for Undergraduate Research at the University at Albany. Project titled: Deep Learning Based Optical Camera Communications. Co-author of two publications: OSA Advanced Photonics Congress 2019 and IEEE Communications Letters 2020.

PUBLICATIONS

MANUSCRIPTS UNDER REVIEW

Saksham Dewan, Hany Elgala, Mohammed Elamassie, and Murat Uysal (2025). Toward AI-Native Physical Layer Design in Optical Wireless Communications. IEEE Standards Magazine. (Under Review).

PEER-REVIEWED JOURNAL ARTICLES

1. D. N. Anwar, H. Bany Salameh, M. Ayyash, H. Elgala and M. Umar, "Efficient DRL-Based 3D UAV Deployment for Optimizing Mobile User Monitoring Under Wireless Service-Reliability Constraints, " in IEEE Open Journal of Vehicular Technology, vol. 7, 1948-1963, 2026.
2. Eltokhey, Mahmoud Wafik, Muhammad Salman Bashir, Othman Isam Younus, Zabih Ghassemloooy, Harald Haas, George K. Karagiannidis, Shivani Rajendra Teli et al. "Optical wireless communications: Illuminating the path for Cooper's law." IEEE Open Journal of the Communications Society (2026).
3. Singh, A., Salameh, H. B., Ayyash, M., & Elgala, H. (2025). Optimal UAV deployment and LED power allocation strategies for VLC-enabled indoor communication networks. Physical Communication, 73, 102884.
4. Saksham Dewan, Hany Elgala, Sami Muhaidat, and Paschalis Sofotasios, "Leveraging multi-task learning to model task relationships in composite VLC waveform demodulation, " Appl. Opt. 64, 8716-8721 (2025).
5. Singh, G., Salameh, H. B., Ayyash, M., Al-Obiedollah, H., & Elgala, H. (2025). On the performance of hybrid TDMA-NOMA-RIS empowered indoor flying networks under dynamic blockages and imperfect SIC. Physical Communication, 71, Article 102691.
6. Singh, Anand, Haythem Bany Salameh, Moussa Ayyash, and Hany Elgala. "Energy Efficient OIRS-aided VLC Systems Employing ML-based User Orientation and Obstacle Awareness." IEEE Sensors Journal (2024).
7. Dewan, Saksham, and Hany Elgala. "Towards Low Complexity VLC Systems: A Multi-Task Learning Approach." Comm&Optics Connect 1, no. 1 (2024): 1-10.
8. Singh, Anand, Haythem Bany Salameh, Moussa Ayyash, and Hany Elgala. "Performance Analysis of OIRS-Aided Indoor VLC Systems Under Dynamic Human Blockages and Random UE Orientation." IEEE Internet of Things Journal (2024).
9. Ahmad, Rizwana, Haythem Bany Salameh, Hany Elgala, Moussa Ayyash, Sufyan Almajali, and Reyad El-Khazali. "Generalized Hybrid LiFi-Wi-Fi UniPHY Learning Framework Towards Intelligent UAV-based Indoor Networks." International Journal of Intelligent Networks (2024).
10. Shrestha, Selina, Shima Naser, Lina Bariah, Sami Muhaidat, Paschalis C. Sofotasios, Hany Elgala, and Ernesto Damiani. "Autoencoder-Based Spatial Modulation for the Next Generation of Wireless Networks." IEEE Internet of Things Journal (2024).
11. Singh, Anand, Haythem Bany Salameh, Moussa Ayyash, and Hany Elgala. "Efficient Power Allocation and Saving Framework for VLC-enabled Indoor Networks with Crowded Heterogeneous Obstacles." IEEE Sensors Journal (2024).
12. Dil Nashin Anwar, Rizwana Ahmad, Haythem Bany Salameh, Hany Elgala, and Moussa Ayyash. Performance Analysis of Neural Network-based Unified Physical Layer for Indoor Hybrid LiFi-Wi-Fi Flying Networks. Neural Computing and Applications, Springer London, pages 1-11, 2023.
13. Rizwana Ahmad, Hany Elgala, Sufyan Almajali, Haythem Bany Salameh, and Moussa Ayyash. Unified Physical Layer Learning Framework towards VLC-enabled 6G-Indoor-Flying-Networks. IEEE Internet of Things Journal, 2023.
14. Mohannad Alhafnawi, Haythem Bany Salameh, Ala'Eddin Masadeh, Haitham Al-Obiedollah, Moussa Ayyash, Reyad El-Khazali, and Hany Elgala. A Survey of Indoor and Outdoor UAV-based Target Tracking Systems: Current Status, Challenges, Technologies, and Future Directions. IEEE Access, 2023.
15. Khadr, M., Bany Salameh, H., Ayyash, M., Elgala, H., & Almajali, S. Jamming Resilient Multi-Channel Transmission for Cognitive Radio IoT-based Medical Networks. IEEE Journal of Communications and Networks, 24(6): 666-678, 2022.
16. Lina Bariah, Wael Jaafar, Sami Muhaidat, Hany Elgala, and Halim Yanikomeroglu. On the Error Performance of LoRa-enabled Aerial Networks over Shadowed Rician Fading Channels. IEEE Communications Letters, 26(10): 2322-2326, 2022.
17. Bushra Alhijawi, Sufyan Almajali, Hany Elgala, Haythem Bany Salameh, and Moussa Ayyash. A Survey on DoS/DDoS Mitigation Techniques in SDNS: Classification, Comparison, Solutions, Testing Tools and Datasets. Computers and Electrical Engineering, ELSEVIER, 99: 107706, 2022.
18. Haythem A Bany Salameh, Monette H Khadr, Mohammad Al-Quraan, Moussa Ayyash, Hany Elgala, and Sufyan Almajali. Jamming-Aware Simultaneous Multi-Channel Decisions for Opportunistic Access in Delay-Critical IoT-Based Sensor Networks. IEEE Sensors Journal, 22(3): 2889-2898, 2021.

19. Ahmed F Hussein, Dola Saha, and Hany Elgala. Mixed-Carrier Communication for Technology Division Multiplexing. *Electronics*, MDPI, 10(18): 2248, 2021 (This article belongs to the Special Issue Optical Wireless Communications and Networking: A Pathway towards 6G).
20. Safaa Hriez, Sufyan Almajali, Hany Elgala, Moussa Ayyash, and Haythem Bany Salameh. A Novel Trust-Aware and Energy-Aware Clustering Method that Uses Stochastic Fractal Search in IoT-Enabled Wireless Sensor Networks. *IEEE Systems Journal*, 16(2): 2693-2704, 2021.
21. Monette H Khadr, Hany Elgala, Michael Rahaim, Abdallah Khreishah, Moussa Ayyash, and Thomas Little. Machine Learning-Based Security-Aware Spatial Modulation for Heterogeneous Radio-Optical Networks. *Proceedings of the Royal Society A*, 477(2248): 20200889, 2021.
22. Monette H Khadr, Ian Walter, Hany Elgala, and Sami Muhaidat. Machine Learning-Based Massive Augmented Spatial Modulation (ASM) for IoT VLC Systems. *IEEE Communications Letters*, 25(2): 494-498, 2020.
23. Monette H Khadr and Hany Elgala. Augmented Communications: Spectral Efficiency and Security Enhanced Visible Light Communications by Design. *Chinese Optics Letters*, 18(9): 090601, 2020.
24. Ahmed F Hussein and Hany Elgala. Wireless Link Pairing toward Secured 6G Networks. *Optics Letters*, 45(14): 4005-4008, 2020.
25. Ahmed F Hussein and Hany Elgala. Design and Spectral Analysis of Mixed-Carrier Communication for Sixth-Generation Networks. *Proceedings of the Royal Society A*, 476(2238): 20200165, 2020.
26. Ahmed Al Hammadi, Paschalis C Sofotasios, Sami Muhaidat, Mahmoud Al-Qutayri, and Hany Elgala. Non-Orthogonal Multiple Access for Hybrid VLC-RF Networks with Imperfect Channel State Information. *IEEE Transactions on Vehicular Technology*, 70(1): 398-411, 2020.
27. Ha-Vu Tran, Georges Kaddoum, Hany Elgala, Chadi Abou-Rjeily, and Hemani Kaushal. Light-Wave Power Transfer for Federated Learning-Based Wireless Networks. *IEEE Communications Letters*, 24(7): 1472-1476, 2020.
28. Elgala, H., Alouini, M. S., Haas, H., Rahaim, M., Tabassum, H., & Watanabe, T. (2019). Introduction to the Special Section on Coexisting Radio and Optical Wireless Deployments (CROWD). *IEEE Transactions on Cognitive Communications and Networking*, 5(4), 1178-1181.
29. Shao, S., Liu, G., Khreishah, A., Ayyash, M., Elgala, H., Little, T. D. C., & Rahaim, M. (2019). Optimizing Handover Parameters by Q-learning for Heterogeneous Radio-Optical Networks. *IEEE Photonics Journal*, 12 (1), 1-15.
30. Almajali, S., El Diehn, D. I., Bany Salameh, H., Ayyash, M., & Elgala, H. (2019). A distributed multi-layer MEC-cloud architecture for processing large scale IoT-based multimedia applications. *Multimedia Tools and Applications*, 78(17), 24617-24638.
31. Rahaim, M., Abdalla, I., Ayyash, M., Elgala, H., Khreishah, A., Little, T. D. C. (2019): Welcome to the CROWD: Design Decisions for Coexisting Radio and Optical Wireless Deployments: *IEEE Network - The Magazine of Global Internet working*, 33(5), 174-182.
32. Mazahir, S., Chaaban, A., Elgala, H., & Alouini, M. S. (2019). Achievable Rates of Multi-Carrier Modulation Schemes for Bandlimited IM/DD Systems. *IEEE Transactions on Wireless Communications*, 18(3), 1957-1973.
33. Nauryzbayev, G., Abdallah, M., & Elgala, H. (2018). Outage of SEE-OFDM VLC-NOMA Networks. *IEEE Photonics Technology Letters*, 31(2), 121-124.
34. Chaaban, A., Chi, N., Elgala, H., Rezki, Z., Safari, M., Sandalidis, H. G., & Imran, M. A. (2018). *IEEE Access Special Section Editorial: Optical Wireless Technologies for 5G Communications and Beyond*. *IEEE Access*, 6, 63619-63623.
35. Bany Salameh, H., Almajali, S., Ayyash, M., & Elgala, H. (2018). Batch-based Security-aware Spectrum Sharing with Simultaneous Assignment Decisions in Time-critical IoT Networks with Cognitive Radio Capabilities. *Transactions on Emerging Telecommunications Technologies*, 29(11), e3317.
36. Khreishah, A., Shao, S., Gharaibeh, A., Ayyash, M., Elgala, H., & Ansari, N. (2018): A Hybrid RF-VLC System for Energy Efficient Wireless Access: in the *IEEE Transactions on Green Communications and Networking*, 2(4), 932-944.
37. Bany Salameh, H., Almajali, S., Ayyash, M., & Elgala, H. (2018): Spectrum Assignment in Cognitive Radio Networks for Internet-of-Things Delay-Sensitive Applications under Jamming Attacks: in the *IEEE Internet of Things Journal*, 5(3), 1904-1913.
38. Shao, S., Khreishah, A., & Elgala, H.: Pixelated VLC-backscattering for Self-Charging Indoor IoT Devices: *IEEE Photonics Technology Letters*: 29(2): 177-180.
39. Ayyash, M., Elgala, H., Khreishah, A., Jungnickel, V., Little, T. D. C., Shao, S., Rahaim, M., Schulz, D., Hilt, J., & Freund R. (2016): Coexistence of Wi-Fi and LiFi Toward 5G: Concepts, Opportunities, and Challenges: *IEEE Communications Magazine*, 54(2): 64-71.
40. Sihua S., Khreishah, A., Ayyash, M., Rahaim, M., Elgala, H., Jungnickel, V., Schulz, D., Little, T. D. C., Hilt, J., & Freund, R. (2015). Design and Analysis of a Visible Light-Communication Enhanced Wi-Fi System. *Journal of Optical Communications and Networking*, 7(10), 960-973.
41. Gancarz, J. E., Elgala, H., & Little, T. D. C. (2015). Overlapping PPM for Band-limited Visible Light Communication and Dimming. *Journal of Solid-State Lighting*, 2(1), 3.
42. Elgala, H., & Little, T. D. C. (2015). Polar-based OFDM and SC-FDE links Toward Energy-Efficient Gbps Transmission under IM-DD Optical System Constraints. *IEEE/OSA Journal of Optical Communications and Networking*, 7(2), A277-A284 (Invited).
43. Butala, P., Elgala, H., & Little, T. D. C. (2014). Sample Indexed Spatial Orthogonal Frequency Division Multiplexing. *Chinese Optics Letters*, 12(9), 090602.

44. Gancarz, J. E., Elgala, H., & Little, T. D. C. (2013). Impact of Lighting Requirements on VLC Systems. *IEEE Communications Magazine*, 51(12), 34-41.
45. Elgala, H., & Little, T. D. C. (2013). Reverse Polarity Optical-OFDM (RPO-OFDM): Dimming Compatible OFDM for Gigabit VLC links. *Optics Express*, 21(20), 24288-24299.
46. Mesleh, R., Elgala, H., & Haas, H. (2012). LED Nonlinearity Mitigation Techniques in Optical Wireless OFDM Communication Systems. *Journal of Optical Communications and Networking*, 4(11), 865-875.
47. Elgala, H., Mesleh, R., & Haas, H. (2011). Indoor Optical Wireless Communication: Potential and State of the Art. *IEEE Communications Magazine*, 49(9), 56-62.
48. Mesleh, R., Elgala, H., & Haas, H. (2011). On the performance of different OFDM based optical wireless communication systems. *IEEE/OSA Journal of Optical Communications and Networking*, 3(8), 620-628.
49. Mesleh, R., Elgala, H., & Haas, H. (2011). Optical Spatial Modulation. *IEEE/OSA Journal of Optical Communications and Networking*, 3(3), 234-244.
50. Mesleh, R., Elgala, H., Mehmood, R., & Haas, H. (2010). Performance of Optical Spatial Modulation with Transmitters-Receiver Alignment. *IEEE Communications Letters*, 15(1), 79-81.
51. Elgala, H., Mesleh, R., & Haas, H. (2010). An LED Model for Intensity-modulated Optical Communication Systems. *IEEE Photonics Technology Letters*, 22(11), 835-837.
52. Elgala, H., Mesleh, R., & Haas, H. (2009). Indoor Broadcasting via White LEDs and OFDM. *IEEE Transactions on Consumer Electronics*, 55(3), 1127-1134.
53. Elgala, H., Mesleh, R., & Haas, H. (2009). Non-linearity Effects and Predistortion in Optical OFDM Wireless Transmission using LEDs. *International Journal of Ultra-Wideband Communications and Systems*, 1(2), 143-150 (Invited).

PEER-REVIEWED CONFERENCE PROCEEDINGS

54. Saksham Dewan and Hany Elgala, End-to-End Learned Non-Power-of-Two Constellations for Adaptive VLC Transmission, *IEEE International Conference on Machine Learning for Communication and Networking (ICMLCN)*, Abu Dhabi, United Arab Emirates, June 8–11, 2026.
55. Honglan Chen, Hany Elgala, Saksham Dewan and Siyuan Cheng, Optical OFDM Modulation Identification with Deep Learning, *IEEE International Conference on Machine Learning for Communication and Networking (ICMLCN)*, Abu Dhabi, United Arab Emirates, June 8–11, 2026.
56. B. Baloch, H. Elgala and S. Dewan, "Enabling RF Backscattering via Dual-Purpose LiFi Transmitter Design: A Harmonic Extraction Approach," *2025 IEEE Middle East Conference on Communications and Networking (MECOM)*, Cairo, Egypt, 2025, pp. 1-6.
57. A. Singh, H. B. Salameh, M. Ayyash and H. Elgala, "IRS-Assisted VLC Versus mmWave Indoor Communication Systems: Performance Comparison and Trade-Offs," *2025 International Conference on Intelligent Computing, Communication, Networking and Services (ICCNS)*, Varna, Bulgaria, 2025, pp. 54-59.
58. Elgala, H., & Stanczak, S. Single Block Index Modulation OFDM (SBIM-OFDM). *IEEE Middle East Conference on Communications and Networking*, November 17–20, 2024, Abu Dhabi, United Arab Emirates, MECOM 2024 [Best Paper Award].
59. Mohannad Alhafnawi, Ala'eddin Masadeh, Haythem Bany Salameh, Moussa Ayyash, Reyad El-Khazali, and Hany Elgala. A Review of Indoor UAV-Based Tracking Systems: Classification, Status, and Challenges. In *2022 International Arab Conference on Information Technology (ACIT)*, IEEE, 2022, pp. 1–6.
60. Khadr, M., & Elgala, H. (2021). Naive Bayesian-Based Augmented Spatial Modulation (ASM) for VLC Systems. In *Signal Processing in Photonic Communications*. Optica July 26–30, 2021.
61. Khadr, M., Ross, Anya, & Elgala, H. (2021). Autoencoder Model for OFDM-Based Optical Wireless Communications Systems Employing Augmented Communications. In *Signal Processing in Photonic Communications*. Optica July 26–30, 2021.
62. Khadr, M., Elgala, H., Ayyash, M., Almajali, S., & Bany Salameh, H. (2020). Testbed Validation of Security-Aware Channel Assignment in IoT-based Cognitive Radio Networks *IEEE World Forum on Internet of Things (WF-IoT)*, doi: 10.1109/WF-IoT48130.2020.9221302, pp. 1–6.
63. Khadr, M., Bany Salameh, H., Ayyash, M., Almajali, S., & Elgala, H. (2019). Securing IoT Delay-Sensitive Communications with Opportunistic Parallel Transmission Capability. *IEEE Global Communications Conference (GLOBECOM)*, pp. 1–6.
64. Khadr, M., Elgala, H., Ayyash, M., Little, T. D. C., Rahaim, M., & Khreishah, A. (2019). Technology Independent Security Aware OFDM (SA-OFDM). *IEEE 30th Annual International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)*, pp. 1–6.
65. Khadr, M. and H. Elgala (2019). Augmented Spatial Modulation (ASM): Secure and Efficient IoT Compatible MIMO Based VLC Links. *IEEE 30th Annual International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)*, pp. 1-6.
66. Pachpande, P., Khadr, M., Hussein, A., Elgala, H., & Saha, Dola (2019). Autoencoder Model for OFDM-based Optical Wireless Communication. In *Signal Processing in Photonic Communications*. Optica, SpT2E-3 pp.
67. Walter, I., Khadr, M., & Elgala, H. (2019). Deep Learning Based Optical Camera Communications. In *Signal Processing in Photonic Communications*. Optica, SpT2E-3 pp.

68. Almajali, S., Bany Salameh, H., Ayyash, M., & Elgala, H. (2019). Extending NS3 to Simulate Cognitive Radio Wireless Networks in a Jammed Environment. 6th International Conference on Software Defined Systems (SDS), pp. 118–122.
69. Shao, S., Khan, Z., Liu, G., Khreishah, A., Ayyash, M., Elgala, H., & Rahaim, M. (2019). Optimizing Handover Parameters by Q-learning for Heterogeneous RF-VLC Networks. IEEE INFOCOM 2019-IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS), pp. 1069–1070.
70. Hussein, A., Elgala, H., & Little, T. D. C. (2018). Evolution of Multi-Tier Transmission Towards 5G LiFi Networks. IEEE Global Communications Conference (GLOBECOM), pp. 1–7.
71. Pachpande, P., Khadr, M., Hussein, A., & Elgala, H. (2018). Visible Light Communication Using Deep Learning Techniques. IEEE 39th Sarnoff Symposium, pp. 1–6.
72. Khadr, M., Elgala, H., Ayyash, M., Little, T. D. C., Khreishah, A., & Rahaim, M. (2018). Security Aware Spatial Modulation (SA-SM). IEEE 39th Sarnoff Symposium. pp. 1–6.
73. Careem, M., Khadr, M., Hussein, A., Saha, D., Elgala, H., & Dutta, A. (2018). CHRONOS: A Cloud-based hybrid RF-Optical Network Over Synchronous Links. IEEE 5G World Forum (5GWF), 351-356 pp.
74. Li, Z., Shao, S., Khreishah, A., Ayyash, M., Iman, A., Elgala, H., Rahaim, M., & Little, T. D. C. (2018). Design and Implementation of a Hybrid RF-VLC System with Bandwidth Aggregation. IEEE 14th International Wireless Communications & Mobile Computing Conference (IWCMC), 194-200 pp.
75. Naurzybayev, G., Abdallah, M., & Elgala, H. (2018). On the performance of NOMA-enabled spectrally and energy efficient OFDM (SEE-OFDM) for Indoor Visible Light Communications. IEEE 87th Vehicular Technology Conference (VTC Spring), pp. 1–5.
76. Bany Salameh, H., Almajali, S., Ayyash, M., & Elgala, H. (2018). Securing Delay-sensitive Cognitive Radio IoT Communications Under Reactive Jamming Attacks: Spectrum Assignment Perspective. Fifth International Conference on Software Defined Systems (SDS), 20-24 pp.
77. Almajali, S., Bany Salameh, H., Ayyash, M., & Elgala, H. (2018). A Framework for Efficient and Secured Mobility of IoT Devices in Mobile Edge Computing. Third international conference on fog and mobile edge computing (FMEC), 58-62 pp.
78. Zhang, Z., Chaaban, A., Shen, C., Elgala, H., Ng, T. K., Boon, S. O., & Alouini, M. S. (2018). Worst-case Residual Clipping Noise Power Model for Bit Loading in LACO-OFDM. Global LIFI Congress (GLC), pp. 1–6.
79. Hussein, A. & Elgala, H. (2018). Lightweight Multi-carrier Modulation for IoT. In Broadband Access Communication Technologies XII. International Society for Optics and Photonics, 10559, p.105590W.
80. Hussein, A., Elgala, H., & Little, T. D. C. (2018). Visible Light Communications: Toward multi-service waveforms. 15th IEEE Annual Consumer Communications & Networking Conference (CCNC), pp. 1–6.
81. Mazahir, S., Chaaban, A., Elgala, H., & Alouini, M. S. (2017). Effective Information Rates of Single-carrier and Multi-carrier Modulation Schemes for Bandwidth Constrained IM/DD Systems. IEEE International Conference on Communications (ICC), pp. 1–6.
82. Bany Salameh, H., Almajali, S., Ayyash, M., & Elgala, H. (2017). Security-aware Channel Assignment in IoT-based Cognitive Radio Networks for Time-critical Applications. Fourth International Conference on Software Defined Systems (SDS), 43-47 pp.
83. Hussein, A., Elgala, H., Fahs, B., & Hella, M. (2017). Experimental Investigation of DCO-OFDM Adaptive Loading using Si PN-based Receiver. 26th Wireless and Optical Communication Conference (WOCC), pp. 1–5.
84. Rahaim, M., Little, T. D. C., Elgala, H., & Govindasamy, S. (2017). An Ultra-Dense IoT Architecture using Hybrid CSMA with Sector Based Scheduling (CSMA/SS) via Visible Light Communications. International Conference on Embedded Wireless Systems and Networks, 335-336 pp.
85. Ali, R. & Elgala, H. (2016). Energy Efficient LED Layout Optimization for Near-uniform Illumination. 15th International Conference on Solid State Lighting and LED-based Illumination Systems. International Society for Optics and Photonics, 9954, p. 995406.
86. Butala, P., Elgala, H., & Little, T. D. C. (2015). Performance of Color Shift Keying under Non-Linear System Model and Illumination Constraints. IEEE GLOBECOM Workshops, pp. 1–7.
87. Mirvakili, A., Koomson, V. J., Rahaim, M., Elgala, H., & Little, T. D. C. (2015). Wireless Access Testbed through Visible Light and Dimming Compatible OFDM. In 2015 IEEE Wireless Communications and Networking Conference (WCNC), 2268-2272 pp.
88. Elgala, H., Wilson, S. K., & Little, T. D. C. (2015). Optical Polar OFDM: On the Effect of Time-domain Power Allocation under Power and Dynamic-range Constraints. IEEE Wireless Communications and Networking Conference (WCNC), 31-35 pp.
89. Butala, P., Elgala, H., Little, T. D. C., & Zarkesh-Ha, P. (2014). Multi-wavelength Visible Light Communication System Design. IEEE GLOBECOM Workshops, 530-535 pp.
90. Little, T. D. C. & Elgala, H. (2014). Adaptation of OFDM under Visible Light Communications and Illumination Constraints. 48th Asilomar Conference on Signals, Systems and Computers, 1739-1744 pp.
91. Shao, S., Khreishah, A., Rahaim, M., Elgala, H., Ayyash, M., Little, T. D. C., & Wu, J. (2014). An indoor hybrid Wi-Fi-VLC internet access system. IEEE 11th International Conference on Mobile Ad Hoc and Sensor Systems, 569-574 pp.
92. Elgala, H. & Little, T. D. C. (2014). SEE-OFDM: Spectral and Energy Efficient OFDM for Optical IM/DD Systems. IEEE 25th annual international symposium on personal, indoor, and mobile radio communication (PIMRC), 851-855 pp.

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